

Devanshi Patel

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EDUCATION

University of Pennsylvania (UPenn), MSE Electrical and Electronics Engineering	Aug 2025 – May 2027
Digital VLSI Design, Computer Organization & Design, Semiconductor Memory Design, Mixed-Signal Circuits, Analog IC Design	
Pandit Deendayal Energy University	July 2021 – May 2025
B.Tech Information and Communication Technology (Electrical Engineering)	GPA: 3.91 / 4.00

EXPERIENCE

Graduate Research Assistant Miskin Lab, University of Pennsylvania	Feb 2026 – Present
- Studied analog and neuromorphic learning frameworks: contrastive local learning and Equilibrium Propagation for hardware-based ML - Analyzed nonlinear analog networks using MOSFET-based elements and energy-minimization principles for low-power inference - Investigated locally adaptive learning rules and their scalability for implementing efficient on-chip training without global backpropagation	
Graduate Hardware Design Project — RISC-V Chip Tapeout University of Pennsylvania	Feb 2026 – Present
- Generated RISC-V SoC RTL via Chipyard (Chisel); authored SDC constraints for multi-corner synthesis targeting TSMC 40nm PDK - Ran complete mflowgen ASIC flow: logic synthesis (Genus), place-and-route with CTS (Innovus), STA-driven timing closure, and PPA analysis across synthesis iterations - Verified RTL functionality via waveform debugging; resolved setup/hold violations and achieved full tapeout sign-off	
Electronics Research Intern eInfochips	May 2024 – Jun 2024
- Designed, simulated 1-bit Sigma-Delta ADC using MATLAB Simulink, applying noise shaping, feedback techniques for signal conversion - Conducted literature survey of ADC architectures; analyzed chip manufacturing data (>60nm), gaining exposure to ASIC design flow and semiconductor processes	

PROJECTS

Pipelined RISC-V Processor CIS 5710: Computer Organization & Design SystemVerilog (GitHub)	Jan 2026 – May 2026
- Implemented 5-stage pipelined RISC-V processor in SystemVerilog with full RV32I ISA, data forwarding, hazard detection unit, and branch resolution logic for correct pipeline execution - Designed AXI-Lite memory interface and arithmetic units including a carry-lookahead adder (CLA) and integer divider; validated correctness using cocotb testbenches and RISC-V ISA compliance suites - Developed and verified single-cycle and multicycle RISC-V microarchitectures as baseline implementations; automated simulation regression using Tcl scripting across all pipeline stages	
Low-Power CNN Accelerator (FPGA) SystemVerilog	Jan 2025 – April 2025
- Implemented CNN architecture with convolution, pooling, and fully connected layers on Zybo Z7-20 (Vivado) with a pipelined fixed-point datapath for low-power inference - Applied RTL-level clock gating, 12-bit fixed-point quantization, memory reuse to reduce switching activity and optimize resource utilization - Reduced dynamic power by 51.79% and total power by 37.34% while maintaining >98% accuracy with efficient hardware utilization	
6T SRAM Memory Array (16×4) Design and Optimization Cadence Virtuoso (45nm)	Nov 2025 – Dec 2025
- Implemented differential bit-line SRAM ($V_{DD} \approx 1.2$ V); sized transistors for read stability (SNM) and write margin; designed periphery including precharge, tri-state drivers, and NAND-based 4×16 row decoder - Optimized array by centralizing precharge and output drivers, reducing access delay by 40.2%, power by 23.2%, and achieving 54.19% FOM improvement across the full array	
Emerging NVM Characterization: RRAM and STT-MRAM Cadence Virtuoso (Verilog-A)	April 2026
- Integrated validated Verilog-A compact models in Cadence; designed voltage-mode and current-mode sense amplifiers for 1-bit and 32-bit array read/write verification; benchmarked PPA trade-offs to guide NVM selection for AI hardware - Characterized switching delay, read latency, and write energy: STT-MRAM achieved $31\times$ lower write energy at bit-cell level and $158\times$ lower at the 32-bit array level vs. RRAM, with sub-10 ns latencies across both technologies	
8-Bit Pipeline ADC Cadence Virtuoso (AMI 0.6μm)	Feb 2026 – May 2026
- Designed full 8-bit pipelined ADC in 600nm CMOS: bootstrapped track-and-hold (5 MS/s), static latch comparator, switched-capacitor MDAC with stage-scaled capacitors (1–8 pF), and 7-stage shift register for digital output alignment - Validated end-to-end at $f_s = 10$ MHz; extracted DNL ± 0.8 LSB, INL ± 1 LSB, SFDR = 22.6 dB, SNDR = 15.2 dB, ENOB = 3.46 bits via MATLAB FFT; measured 160 mW total power (17.8 mW dynamic); FOM = 1.454 nJ/conversion-step	
Optimized Carry Select Adder (OCSA) Cadence Virtuoso (90nm) (IEEE Publication)	Oct 2024 – Dec 2024
- Designed OCSA using pre-computation and MDCVSL-based multiplexer logic to improve computation speed over conventional CSA - Reduced propagation delay by 10.52% and power by $\approx 20\%$, achieving lower PDP for low-power IoT applications; published at IEEE SETCOM 2025	

SKILLS

Languages & HDL: Verilog, SystemVerilog, VHDL, C/C++, Python, MATLAB, Tcl, Perl, Git, Linux, Verilog-A
Digital / VLSI: RTL Design, FSM & Datapath Design, Logic Synthesis, SDC Constraints, STA, Timing Closure, CTS, PPA Analysis, ASIC Design Flow, AXI-Lite, Clock Gating, Low-Power Design, SRAM Design, Critical Path Optimization
Architecture & Verification: Processor Microarchitecture (Single-Cycle, Multicycle, Pipelined), Datapath Design (ALU, Reg File, Control Unit), FPGA Prototyping, Functional Verification, Testbench Development, RTL Simulation, Fixed-Point Design
EDA Tools: Cadence Virtuoso, Cadence Spectre, Genus, Innovus, Xcelium, Xilinx Vivado, Chipyard, mflowgen, cocotb